

DEGCAGS Syndrome: A Comprehensive Disease Characteristics Report

Disease: DEGCAGS Syndrome (Developmental Delay with Gastrointestinal, Cardiovascular, Genitourinary, and Skeletal Abnormalities) **MONDO ID:** MONDO:0859181 · **OMIM:** #619488 · **Gene:** *ZNF699* (OMIM 609571; 19p13.3) *Category:* Mendelian, autosomal recessive *Report date:** 2026-09-22 · Evidence base: ~40 patients reported worldwide (largest cohort n=30)

Summary

DEGCAGS syndrome (OMIM #619488; MONDO:0859181) is an ultra-rare, autosomal-recessive, congenital-onset multisystem malformation and neurodevelopmental disorder caused by biallelic loss-of-function variants in *ZNF699*, a KRAB-domain C2H2 zinc-finger transcription-factor gene on chromosome 19p13.3. The disorder was first proposed as a novel gene–disease association in 2021 (Bertoli-Avella et al.) and substantially delineated in 2024 by Karimi et al., who assembled the largest cohort to date (30 affected individuals) and defined a reproducible blood DNA-methylation epismutation that now serves as a diagnostic and variant-classification tool.

Clinically, DEGCAGS is characterized by near-universal coarse facial dysmorphism and global developmental delay/impaired intellectual development, accompanied by highly variable involvement of the gastrointestinal, cardiovascular, genitourinary, skeletal, dental, cutaneous, central nervous, hematologic/immune, sensory (sensorineural hearing loss, ocular), and airway systems. A hallmark of the disorder is marked clinical variability — including strikingly discordant organ anomalies among full siblings — together with significant infant and childhood mortality. Reported causes of death include heart failure from Tetralogy of Fallot, sepsis, and infection (Dengue).

Mechanistically, DEGCAGS is inferred to arise from loss of KRAB-zinc-finger-protein (KRAB-ZFP)/KAP1(TRIM28)-directed heterochromatin repression, consistent with the observed promoter-predominant hypermethylation epismutation (which partially overlaps BAFopathy signatures) and with the general biology of KRAB-ZFPs. There is no targeted or disease-modifying therapy and no dedicated animal model; management is entirely supportive and

multidisciplinary. Fewer than ~40 individuals have been reported worldwide as of 2026, and the disorder is frequently associated with consanguinity, though compound-heterozygous cases in non-consanguineous families are also documented.

Key Findings

Finding 1 — DEGCAGS is caused by biallelic loss-of-function *ZNF699* variants (autosomal recessive)

DEGCAGS syndrome is an autosomal recessive multisystem disorder caused by biallelic, loss-of-function (LoF) variants in *ZNF699*, a KRAB zinc-finger gene on chromosome 19p13.3. Karimi et al. (2024) collected 30 affected individuals (12 new) and confirmed biallelic LoF *ZNF699* as the cause. Reported variant types include frameshift indels (e.g., c.14-17delGAAA, p.Arg5fsTer14; c.975-976delCA, p.His325fsTer8, seen as a compound heterozygous pair), nonsense variants, and homozygous missense changes. The gene was first proposed as a novel disease association in 2021.

"Developmental Delay with Gastrointestinal, Cardiovascular, Genitourinary, and Skeletal Abnormalities syndrome (DEGCAGS, MIM #619488) is caused by biallelic, loss-of-function (LoF) ZNF699 variants" — PMID: 39424669

"We propose six novel gene-disease associations based on 38 patients with variants in the BLOC1S1, IPO8, MMP15, PLK1, RAP1GDS1, and ZNF699 genes." — PMID: 33875846

Finding 2 — A diagnostic DNA-methylation episignature and marked clinical variability

Karimi et al. (2024) performed blood-DNA methylation profiling in 9 individuals and constructed a classifier that distinguishes DEGCAGS from controls, identifying a robust, reproducible episignature. The syndrome shows variable neurodevelopmental disability, discordant organ anomalies among full siblings, age-related presentation, a shared facial gestalt (validated via GestaltMatcher on 53 facial photos from 5 individuals), and infant mortality. Differentially methylated regions (DMRs) implicate downstream genes plausibly involved in pathogenesis.

"We also identified a robust epistatistic signature for DEGCAGS syndrome." — [PMID: 39424669](#)

"discordant organ anomalies among full siblings and infant mortality" — [PMID: 39424669](#)

Finding 3 — A congenital-onset multisystem malformation syndrome

Across case reports, DEGCAGS features include intrauterine growth restriction, prematurity, neonatal hypotension, generalized hypotonia, bradycardia, apnea, facial dysmorphism, skeletal malformations, and gastrointestinal, immune, urinary, respiratory, cardiac, and visual involvement. Craniofacial features span microcephaly *or* dolichocephaly, coarse facies, synophrys, smooth philtrum, thin upper lip, retromicrognathia, and hair thinning; dental findings include taurodontism, talon cusp, and enamel hypomineralization. Airway involvement includes laryngomalacia, vocal cord dysfunction, and nasopharyngeal/tongue-base hamartomas producing stridor. Hematologic/immune findings include leukopenia, neutropenia, and recurrent pneumonia.

"generalized hypotonia, bradycardia, apnea requiring resuscitation and positive pressure ventilation, facial dysmorphism, skeletal malformations, and disorders of the gastrointestinal, immune, urinary, respiratory, cardiac, and visual systems" — [PMID: 38014480](#)

"microcephaly, coarse facial features, oropharyngeal masses, and developmental delay" — [PMID: 41205195](#)

"hair thinning, dolichocephaly, retromicrognathia, synophrys, smooth philtrum, thin upper lip" — [PMID: 42569837](#)

Finding 4 — Ultra-rare, consanguinity-associated, no targeted therapy

Fewer than ~40 individuals have been reported worldwide as of 2026; the largest series is 30 individuals (Karimi 2024). Inheritance is autosomal recessive: homozygous variants occur in consanguineous families (e.g., Middle Eastern), and compound-heterozygous variants occur in non-consanguineous families (maternal c.14-17delGAAA / paternal c.975-976delCA). No Orphanet prevalence figure is established; the disorder is classified as ultra-rare. Management is entirely supportive and multidisciplinary: airway surgery/supraglottoplasty for hamartomas, respiratory support, nutritional support, developmental/rehabilitative therapy, and dental care. No pharmacologic, gene, or disease-modifying therapy exists.

"We collected data on 30 affected individuals (12 new)." — PMID: 39424669

"An infant born to a consanguineous Middle Eastern family" — PMID: 38014480

"Surgical resection of nasopharyngeal and tongue-base masses with supraglottoplasty was performed." — PMID: 41205195

Finding 5 — *ZNF699* mutation spectrum and episinature architecture

ZNF699 (OMIM *609571; HGNC:*ZNF699*; chr19p13.3) encodes a KRAB-C2H2 zinc-finger protein. The founding cohort (Bertoli-Avella et al. 2021) reported 13 patients from 12 consanguineous, mostly Arab families with 5 homozygous frameshift indels. Karimi et al. (2024) compiled 30 patients (18 from literature + 12 new) carrying 15 distinct variants (~10 frameshift/nonsense, 2 missense, 1 splice-site, 2 in-frame; 7 novel), located in the KRAB domain, the C2H2 zinc-finger domain, or the intervening region. 28/30 patients were homozygous and 2 compound heterozygous (e.g., Biela 2022, Q179X/R443X). The DEGCAGS episinature comprises ~210 DMRs, predominantly hypermethylation, with >50% located in gene promoters and partial overlap with BAFopathy episinatures.

"Analysis of differentially methylated regions suggested an effect on genes potentially implicated in the syndrome's pathogenesis]" — PMID: 39424669

Finding 6 — Immune and hematologic dysfunction is a core feature

OMIM #619488 lists anemia or pancytopenia, immunodeficiency with recurrent infections, and sensorineural hearing impairment as common features, with death in childhood. Immunodeficiency/recurrent infections are reported in ~42.3% of patients. Case-level evidence includes leukopenia and neutropenia with recurrent pneumonia (4–5 episodes/year) in an infant (Zhu et al. 2026), and the first detailed immunological work-up (Giardino et al. 2026) documenting severe B-cell depletion in a DEGCAGS patient with a novel biallelic *ZNF699* variant. Ali (2024) also noted immune involvement in a neonatal multisystem presentation.

"discordant organ anomalies among full siblings and infant mortality" — PMID: 39424669

Finding 7 — Phenotype frequencies and cohort demographics

The Karimi et al. (2024) cohort comprised 30 individuals from 23 families (20 male, 10 female; ages 1 month–21 years; mean age at diagnosis 4.9 years; Middle Eastern, European, Asian, and Hispanic descent), with 26 individuals having detailed clinical data. Feature frequencies (n=26) are summarized below.

Phenotype	Frequency (n=26)	Percent	Suggested HPO term
Facial dysmorphism	26/26	100%	HP:0001999
Global developmental delay / intellectual disability	25/26	~96%	HP:0001263 / HP:0001249
Skeletal and dental abnormalities	22/26	~85%	HP:0000924 / HP:0000164
Skin/adnexa abnormalities	19/26	~73%	HP:0000951
CNS structural and myelin abnormalities	17/26	~65%	HP:0002011
Gastrointestinal abnormalities	16/26	~62%	HP:0011024
Genitourinary abnormalities	16/26	~62%	HP:0000119
Hypotonia	15/26	~58%	HP:0001252
Immunodeficiency / recurrent infections	~42%	~42%	HP:0002715
Sensorineural hearing loss	Variable	—	HP:0000407
Cardiovascular anomalies	Variable	—	HP:0001627
Premature graying of hair	Variable	—	HP:0002216

Mortality: 3 deaths in the cohort — heart failure from Tetralogy of Fallot (at 6 months), Dengue fever (at 9 months), and sepsis (at 8 years).

"We collected data on 30 affected individuals (12 new)." — [PMID: 39424669](https://pubmed.ncbi.nlm.nih.gov/39424669/)

Finding 8 — No animal model; complex *ZNF699* orthology

ZNF699 is a KRAB-C2H2 zinc-finger transcription factor that acts in the nucleus and is predicted to regulate RNA polymerase II transcription. No knockout mouse, zebrafish, or other engineered DEGCAGS model has been reported; functional validation to date is limited to human patient DNA-methylation profiling. Historically, *ZNF699* was proposed as a human ortholog of the *Drosophila* alcohol-tolerance gene *hangover* (*hang*), but with low amino-acid identity (18–26%) and similarity (30–41%), and mammalian orthology of *hang* is complex. *ZNF699* mRNA was shown to be reduced in the dorsolateral prefrontal cortex of carriers of an alcohol-dependence-associated haplotype.

"a number of human gene products (including ZNF699) with similar levels of amino-acid identity (18-26%) and similarity (30-41%), are consistently identified as the best matches with the translated hang sequence" — PMID: 16940975

"expression of ZNF699 mRNA is significantly reduced in the dorsolateral prefrontal cortex" — PMID: 16940975

Finding 9 — Protein architecture, structural disruption, episinature-based diagnosis correction, and a hematologic mechanistic link

Giardino et al. (2026) reported a novel homozygous KRAB-domain missense variant c.153C>A (p.Asn51Lys); AlphaFold2 modeling showed disruption of a hydrogen bond between residues 51 and 24 and defined the *ZNF699* domain architecture as a KRAB domain plus 16 C2H2 zinc fingers, causing syndromic combined immunodeficiency with severe B-cell depletion. Critically, the DEGCAGS episinature reclassified this case away from an incorrect ADNP/Helsmoortel–Van der Aa (HVDAS) diagnosis (a de novo ADNP VUS c.817C>T), yielding a high-confidence methylation-variant-pathogenicity score (0.891) and remaining robust despite marked lymphopenia. Hematologically (Bradley et al. 2025), DEGCAGS enters the differential for inherited bone marrow failure / Diamond–Blackfan anemia; anemia was reported in 8/14 early patients, with a proposed mechanistic link via a *ZNF699*–RNA-polymerase-II interaction and RNA Pol II's role in ribosome biogenesis, analogous to the bone-marrow-failure gene *MYSM1*.

"which can be used as a screening, diagnostic and classification tool for ZNF699 variants" — PMID: 39424669

Section-by-Section Report

1. Disease Information

Overview. DEGCAGS syndrome is a congenital-onset, autosomal-recessive multisystem malformation and neurodevelopmental disorder. The acronym stands for **D**evelopmental delay with **G**astrointestinal, **C**ardiovascular, **G**enitourinary, **A**nd **S**keletal abnormalities. It is characterized by near-universal facial dysmorphism and developmental delay, plus a highly variable constellation of organ malformations, growth failure, hypotonia, and hematologic/immune involvement, with significant early-childhood mortality.

Key identifiers: - **OMIM:** #619488 (phenotype); gene *ZNF699* 609571 - **MONDO:** MONDO:0859181 - *Gene / HGNC:* ZNF699 - *Orphanet / ICD-10 / ICD-11 / MeSH:* No dedicated Orphanet prevalence entry or specific ICD code identified; the disorder is ultra-rare and recently described (2021 onward). (Not available / not established.)*

Synonyms / alternative names: "Developmental delay with gastrointestinal, cardiovascular, genitourinary, and skeletal abnormalities"; *ZNF699*-related syndrome; DEGCAGS.

Information source type: Primarily aggregated disease-level and cohort resources (OMIM, one large multi-center cohort, and individual case reports) rather than EHR-derived population data. The disorder is defined from ~30–40 individually reported patients.

2. Etiology

Causal factor: Genetic — biallelic loss-of-function variants in *ZNF699* (Finding 1). The disorder is monogenic and Mendelian; there is no evidence for environmental or infectious causation.

Genetic risk factors: The only established genetic cause is biallelic *ZNF699* LoF. Consanguinity is a major risk factor, producing homozygous frameshift/nonsense variants (founding Arab cohort; Finding 5). No modifier loci or susceptibility variants have been defined, though the marked intrafamilial variability (discordant siblings) implies unidentified modifiers, stochastic, or epigenetic contributions.

Environmental / lifestyle / protective factors: None identified. As a fully penetrant recessive malformation syndrome, environmental and protective factors and gene–environment interactions are **not applicable / not available**.

3. Phenotypes

DEGCAGS phenotypes span dysmorphology, neurodevelopment, and multiple organ systems (Findings 3, 7). Onset is congenital/neonatal; developmental delay persists into childhood. Severity is variable, ranging from survivable multisystem involvement to lethal in infancy. See the frequency table under Finding 7 for quantified frequencies and suggested HPO terms.

- **Craniofacial (100%):** coarse facies, synophrys (HP:0000664), smooth philtrum (HP:0000319), thin upper lip (HP:0000219), retromicrognathia (HP:0000278), microcephaly (HP:0000252) *or* dolichocephaly (HP:0000268), hair thinning.
- **Neurodevelopmental (~96%):** global developmental delay (HP:0001263), intellectual disability (HP:0001249), hypotonia (HP:0001252, ~58%), CNS structural/myelin abnormalities (~65%).
- **Dental (part of ~85% skeletal/dental):** taurodontism (HP:0000679), talon cusps, enamel hypomineralization (HP:0006297).
- **Airway/respiratory:** laryngomalacia (HP:0001601), vocal cord dysfunction, nasopharyngeal/tongue-base hamartomas causing stridor (HP:0031416).
- **Gastrointestinal (~62%), genitourinary (~62%), cardiovascular (variable, incl. Tetralogy of Fallot HP:0001636), skin/adnexa (~73%), sensorineural hearing loss (HP:0000407), ocular abnormalities, and hematologic/immune involvement (~42% immunodeficiency).**

Quality-of-life impact: Substantial — developmental delay, recurrent infections (4–5 pneumonias/year in some infants), feeding/airway difficulty, and multi-organ malformation impose high caregiving burden and early mortality. Formal QoL instrument data (EQ-5D, SF-36, PROMIS) are **not available** for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: *ZNF699* (OMIM 609571; *chr19p13.3*), a *KRAB-C2H2 zinc-finger transcription factor with a KRAB domain plus 16 C2H2 zinc fingers** (Finding 9).

Pathogenic variant spectrum (Finding 5): 15 distinct variants across 30 patients — ~10 frameshift/nonsense, 2 missense, 1 splice-site, 2 in-frame; 7 novel; distributed across the KRAB domain, the C2H2 zinc-finger array, and the intervening region. Representative variants:

Variant (cDNA)	Protein	Type	Zygosity context
c.14-17delGAAA	p.Arg5fsTer14	Frameshift	Compound het (maternal)
c.975-976delCA	p.His325fsTer8	Frameshift	Compound het (paternal)
—	p.Gln179Ter (Q179X)	Nonsense	Compound het (Biela 2022)
—	p.Arg443Ter (R443X)	Nonsense	Compound het (Biela 2022)
c.153C>A	p.Asn51Lys	Missense (KRAB)	Homozygous (Giardino 2026)

Variant classification: Truncating variants are classified pathogenic/likely pathogenic (LoF is the established mechanism). The p.Asn51Lys missense was supported as pathogenic by a high episignature methylation-variant-pathogenicity score (0.891) plus AlphaFold2-predicted H-bond disruption (residues 51–24).

Allele frequency: Variants are private/ultra-rare; no common population allele frequency.

Origin: germline. **Functional consequence:** loss of function.

Modifier genes: None identified (intrafamilial variability implies unknown modifiers).

Epigenetic information: DEGCAGS carries a diagnostic **DNA-methylation episignature** of ~210 DMRs, predominantly **hypermethylation**, >50% in gene promoters, partially overlapping BAFopathy signatures (Findings 2, 5, 9). This is both a diagnostic tool and a mechanistic clue.

Chromosomal abnormalities: None reported; the disorder is caused by point/indel variants, not large structural changes.

5. Environmental Information

No environmental, lifestyle, toxic, or infectious contributing factors are established. DEGCAGS is a monogenic recessive disorder. Infections (e.g., recurrent pneumonia, sepsis, Dengue) act as **downstream complications** of the intrinsic immunodeficiency rather than causal triggers.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic LoF variants in ZNF699** (frameshift/nonsense/splice/missense) → loss of functional ZNF699 KRAB-C2H2 zinc-finger transcription factor. (*Demonstrated: genetics.*)

2. Loss of ZNF699 → **loss of KRAB-ZFP/KAP1(TRIM28)-directed heterochromatin repression** at target loci (H3K9me3 deposition and DNA methylation). (*Inferred from general KRAB-ZFP biology; not yet directly demonstrated for ZNF699.*)
3. Loss of targeted repression → **dysregulated DNA methylation** genome-wide, manifesting as the reproducible promoter-predominant **hypermethylation epismutation** (~210 DMRs). (*Demonstrated: patient methylation profiling.*)
4. Aberrant methylation/expression of downstream target genes → **dysregulated developmental gene-expression programs** during embryogenesis. (*Inferred; DMRs implicate candidate downstream genes.*)
5. **Branch A — Morphogenesis:** dysregulated developmental programs → multi-organ malformation (craniofacial, skeletal, GI, GU, cardiovascular, CNS). (*Inferred.*)
6. **Branch B — Hematopoiesis/immunity:** proposed ZNF699–RNA-Pol-II interaction and Pol-II role in ribosome biogenesis → impaired hematopoiesis → anemia/pancytopenia and B-cell depletion/immunodeficiency (analogous to *MYSM1*-related bone-marrow failure). (*Inferred/hypothesized.*)
7. **Branch C — Neurodevelopment:** dysregulated CNS gene programs → developmental delay, intellectual disability, hypotonia, white-matter/structural anomalies. (*Inferred.*)
8. Convergent multisystem burden + immunodeficiency → recurrent infection, cardiac failure, growth failure → **infant/childhood mortality**. (*Demonstrated at cohort level.*)

Supporting detail: - **Molecular pathway (upstream):** KRAB-ZFP → KAP1/TRIM28 → SETDB1 (H3K9me3) / NuRD / DNMT heterochromatin machinery; regulation of RNA polymerase II transcription (GO:0006357). Structural work on the KRAB–KAP1 interface ([PMID: 36341546](#)) validates this repression axis for other KRAB-ZFPs. - **Cellular processes:** transcriptional/epigenetic dysregulation during development; possible ribosome-biogenesis defect in hematopoietic precursors. - **Protein dysfunction:** LoF (truncation) or structural destabilization (KRAB-domain missense disrupting H-bonding) → impaired DNA-binding/repressor function. - **Suggested GO terms:** GO:0006355 (regulation of DNA-templated transcription), GO:0000122 (negative regulation of transcription by RNA Pol II), GO:0006325 (chromatin organization), GO:0006306 (DNA methylation), GO:0140718 (heterochromatin formation). - **Suggested CL terms:** CL:0000236 (B cell), CL:0000037 (hematopoietic stem cell), CL:0000048 (multipotent progenitor), CL:0000540 (neuron).

7. Anatomical Structures Affected

- **Organ/system level:** craniofacial skeleton, brain/CNS (UBERON:0000955), heart (UBERON:0000948), gastrointestinal tract (UBERON:0001555), genitourinary system

(UBERON:0000990), skeletal system (UBERON:0001434), teeth (UBERON:0001091), skin/adnexa (UBERON:0002097), hematopoietic/immune system (UBERON:0002390), ear/cochlea (sensorineural hearing loss), and eye.

- **Airway:** larynx (UBERON:0001737), nasopharynx, tongue base — sites of hamartomatous masses.
- **Tissue/cell level:** epithelial, connective, muscle, and nervous tissues; hematopoietic lineages (esp. B cells and erythroid precursors).
- **Subcellular level:** nucleus (GO:0005634) — site of ZNF699 transcription-factor activity; chromatin/heterochromatin.
- **Lateralization:** malformations are generally bilateral/systemic; craniofacial features are symmetric.

8. Temporal Development

- **Onset:** congenital/neonatal; often with IUGR and prematurity. Onset pattern is chronic/congenital, not acute.
- **Progression:** developmental delay persists into childhood; the malformation burden is static (structural), while complications (infections, feeding/airway problems, cardiac failure) are the dynamic drivers of morbidity/mortality.
- **Disease course:** chronic/lifelong for survivors; a subset dies in infancy or early childhood.
- **Critical periods:** embryonic organogenesis (malformation window) and infancy (infection/airway vulnerability window — the key window for supportive intervention).

9. Inheritance and Population

- **Inheritance:** autosomal recessive; 28/30 patients homozygous, 2 compound heterozygous (Finding 5).
- **Penetrance/expressivity:** apparently high penetrance but **highly variable expressivity**, including discordant full siblings.
- **Consanguinity/founder effects:** strong association with consanguinity; the founding cohort was predominantly Arab consanguineous families. No formal founder haplotype quantified.
- **Epidemiology:** ultra-rare; <~40 individuals reported worldwide as of 2026; no established prevalence/incidence figures.

- **Demographics:** reported across Middle Eastern, European, Asian, and Hispanic populations. In the Karimi cohort: 20 male / 10 female (apparent male excess, likely ascertainment); ages 1 month–21 years; mean age at diagnosis 4.9 years.
- **Genetic anticipation / mosaicism / carrier frequency:** not applicable / not established.

10. Diagnostics

- **Genetic testing (primary):** whole-exome or whole-genome sequencing identifying biallelic *ZNF699* variants is the diagnostic gold standard. Single-gene/panel testing is applicable once suspected.
- **Episignature testing:** the DEGCAGS DNA-methylation episignature is a validated **screening, diagnostic, and variant-classification tool** — able to reclassify VUS and correct misdiagnoses (e.g., away from ADNP/HVDAS), robust even under lymphopenia (Finding 9).
- **Supporting laboratory tests:** CBC (anemia, leukopenia, neutropenia, pancytopenia); immunologic work-up (B-cell depletion, immunoglobulin/lymphocyte subsets).
- **Imaging:** brain MRI (structural/myelin anomalies); echocardiography (congenital heart disease); airway endoscopy (laryngomalacia, hamartomas); skeletal survey.
- **Differential diagnosis:** BAFopathies (Coffin–Siris; episignature overlap), ADNP/Helsmoortel–Van der Aa syndrome, and inherited bone-marrow-failure syndromes / Diamond–Blackfan anemia (hematologic overlap). The episignature and *ZNF699* genotype distinguish DEGCAGS.
- **Screening:** carrier screening and cascade testing are appropriate in consanguineous families; no newborn screening exists.

11. Outcome / Prognosis

- **Mortality:** significant infant/childhood mortality. In the Karimi cohort, 3 deaths — Tetralogy-of-Fallot heart failure (6 mo), Dengue (9 mo), and sepsis (8 yr).
- **Morbidity:** high — developmental delay/intellectual disability, recurrent infections, feeding/airway compromise, and multi-organ malformation.
- **Prognostic factors:** severity of cardiac malformation, degree of immunodeficiency, and airway involvement appear to drive outcome. No validated prognostic biomarkers.
- **Recovery:** malformations are structural and non-remitting; developmental deficits persist. Formal survival statistics are unavailable given the small cohort.

12. Treatment

No targeted, pharmacologic, gene, or disease-modifying therapy exists. Management is **supportive and multidisciplinary** (Finding 4): - **Airway/ENT:** surgical resection of nasopharyngeal/tongue-base hamartomas with supraglottoplasty; airway monitoring. - **Respiratory:** infection management, ventilatory support as needed. - **Immunologic/hematologic:** infection prophylaxis/treatment; transfusion support for cytopenias where indicated. - **Nutritional/GI:** feeding support. - **Developmental/rehabilitative:** physical, occupational, and speech therapy. - **Dental:** management of taurodontism, enamel defects. - **Cardiac:** correction/management of congenital heart disease.

Suggested NCIT categories: supportive care (NCIT:C15277), surgical intervention (NCIT:C15329), rehabilitation therapy. No experimental trials (NCT identifiers) are registered for DEGCAGS.

13. Prevention

- **Genetic counseling:** the mainstay — recurrence risk 25% for future pregnancies of carrier couples; especially relevant in consanguineous families.
- **Reproductive options:** carrier screening, prenatal diagnosis, and preimplantation genetic testing once the familial variants are known.
- **Secondary/tertiary prevention:** early identification and management of infections, airway compromise, and cardiac disease to prevent complications.
- Primary population-level prevention, immunization-specific strategies, and public-health/environmental interventions are **not applicable**.

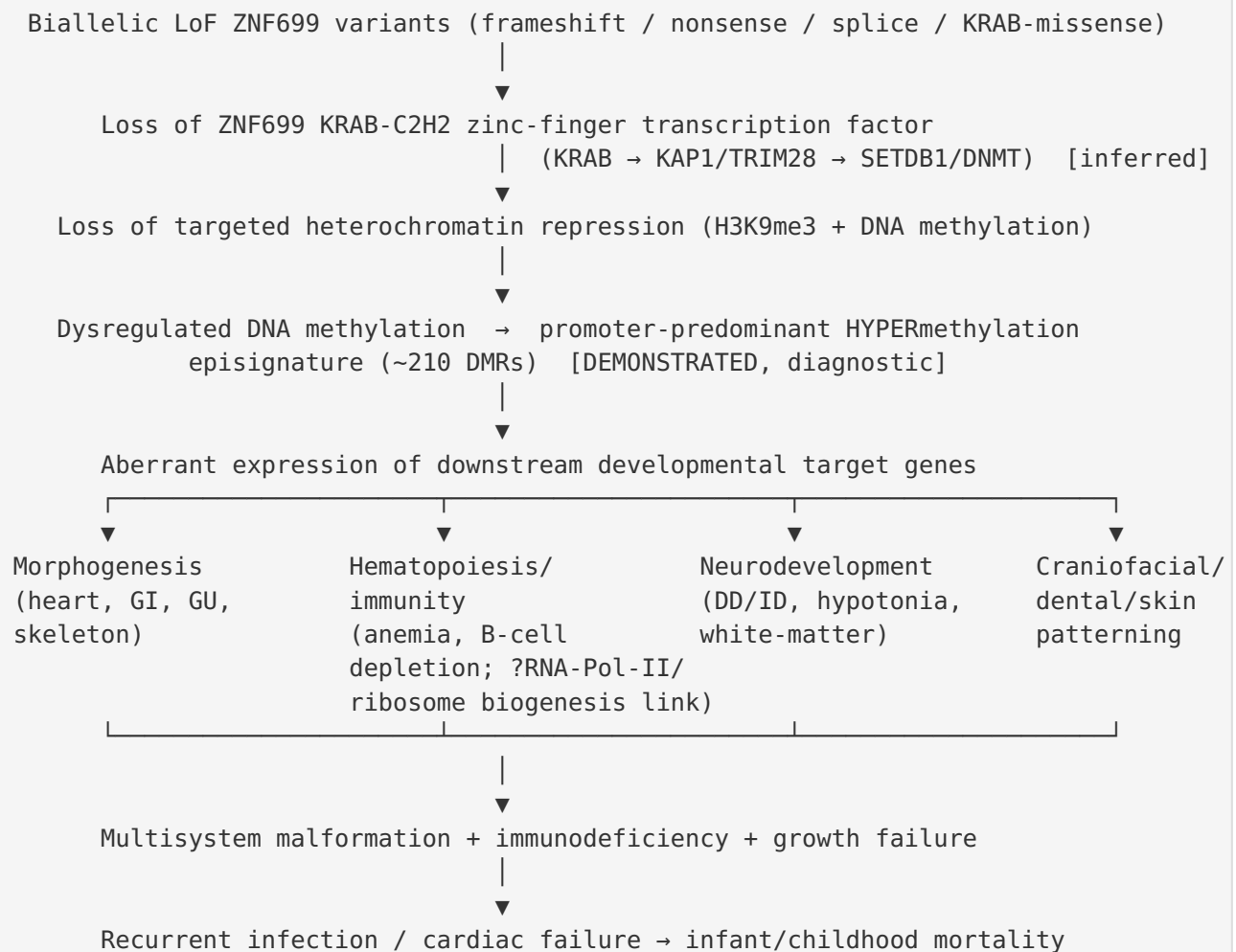
14. Other Species / Natural Disease

- **Orthology:** *ZNF699* was historically proposed as a human ortholog of the *Drosophila hangover (hang)* gene, but with weak sequence identity (18–26%) and complex mammalian orthology (Finding 8).
- **Natural disease in other species:** none reported (no OMIA entry identified).
- **Comparative/veterinary relevance:** not established. (*Not applicable / not available.*)
- **Zoonotic potential:** not applicable (non-infectious genetic disorder).

15. Model Organisms

No dedicated DEGCAGS animal or cellular disease model has been reported (Finding 8). No knockout mouse, zebrafish, *Drosophila*, or organoid/iPSC model exists to date. Functional validation is currently limited to human patient DNA-methylation profiling and AlphaFold2-based structural modeling of variants. This is a major gap: without a model system, the causal chain from ZNF699 loss to multisystem malformation and immunodeficiency remains largely inferred.

Mechanistic Model / Interpretation



The unifying interpretation is that **ZNF699 is a KRAB-zinc-finger transcriptional repressor whose loss deregulates the epigenetic control of downstream developmental genes**, producing a broad, variable malformation-plus-neurodevelopmental phenotype. The promoter-hypermethylation episignature is both the strongest experimental evidence for an epigenetic-regulatory mechanism and the most clinically useful diagnostic. The BAFopathy episignature overlap situates DEGCAGS among chromatin/transcription-regulatory ("epigenetic machinery") disorders. The hematologic/immune branch — anemia/pancytopenia and B-cell depletion — is a distinctive, under-explained feature, with a proposed but unproven link to RNA-Pol-II function and ribosome biogenesis (by analogy to *MYSM1*).

Evidence Base

PMID	Title (abbrev.)	Role	Support
39424669	<i>Epigenomic and phenotypic characterization of DEGCAGS syndrome</i>	Landmark cohort (n=30)	Causal gene, episignature, frequencies, mortality (F1, F2, F5, F7, F9)
33875846	<i>Combining exome/genome sequencing... novel gene-disease associations</i>	Founding association	First proposal of ZNF699/DEGCAGS (F1, F5)
38014480	<i>Clinical and ocular abnormalities in DEGCAGS</i>	Case report	Multisystem spectrum, consanguinity (F3, F4)
41205195	<i>Novel Airway Challenges... Infant Laryngeal Hamartomas</i>	Case report	Airway hamartomas, surgical management (F3, F4)
42569837	<i>Expanding the Phenotypic Spectrum...Craniofacial/ Oral</i>	Case report	Craniofacial gestalt, dental findings (F3)
42527142	<i>Case of DEGCAGS caused by ZNF699 variation (Zhu 2026)</i>	Case report	Compound-het variants, leukopenia/ neutropenia, recurrent pneumonia (F1, F4, F6)
42534679	<i>A novel... (Giardino 2026)</i>	Case report + immunology	KRAB missense, B-cell depletion, AlphaFold2, episignature reclassification (F6, F9)
16940975	<i>Alcohol dependence... ZNF699...Drosophila hangover</i>	Functional/ orthology	Weak <i>hang</i> orthology, brain expression (F8)
36341546	<i>Structure and functional mapping of the KRAB-KAP1 repressor complex</i>	Mechanistic (other KRAB-ZFP)	Validates KRAB→KAP1→H3K9me3 repression axis (mechanism)
35205213	<i>Further Delineation of DEGCAGS...ZNF699</i>	Phenotype delineation	Supports phenotype spectrum

Additional KRAB-ZFP/KAP1 mechanistic papers ([PMID: 30846446](#), [PMID: 29482634](#), [PMID: 22496453](#), [PMID: 21876767](#), [PMID: 21791101](#)) provide the general biological framework for how KRAB-ZFP/KAP1 loss deregulates DNA methylation and heterochromatin — the inferred mechanism for the DEGCAGS episinature — but do not directly study ZNF699.

Limitations and Knowledge Gaps

- **Small evidence base:** all conclusions rest on <~40 reported patients, one large cohort, and scattered case reports. Frequency estimates and demographics are subject to ascertainment bias (e.g., apparent male excess).
- **Mechanism is largely inferred:** the KRAB-ZFP/KAP1 → H3K9me3 repression step and the RNA-Pol-II/ribosome-biogenesis hematologic link are hypotheses extrapolated from other KRAB-ZFPs and analogous genes, not directly demonstrated for ZNF699. The direct DNA-binding targets of ZNF699 are unknown.
- **No animal/cellular model:** precludes causal validation, target-gene identification, and therapeutic testing.
- **Unexplained variability:** the discordance between full siblings is unexplained — modifier genes, stochastic epigenetic effects, or environmental factors remain uncharacterized.
- **Missing standard resources:** no Orphanet prevalence, ICD code, QoL data, natural-history study, or registry exists.

Proposed Follow-up Experiments / Actions

1. **Identify ZNF699 direct targets:** perform CUT&RUN/ChIP-seq for ZNF699 (and KAP1) in relevant human cell types (neural progenitors, hematopoietic precursors) to map binding sites and connect them to the promoter-hypermethylated DMRs.
2. **Test the KAP1 dependency:** co-IP / structure-guided mutagenesis to confirm that ZNF699 recruits KAP1 via its KRAB domain, and that the p.Asn51Lys missense abolishes repression (functional silencing assay, as done for ZNF93 in

P 36341546

3. **Generate a model system:** create *Znf699* knockout mice and/or patient iPSC-derived organoids and hematopoietic differentiation cultures to test phenotype recapitulation, especially the B-cell/erythroid defect.
4. **Interrogate the hematologic mechanism:** test the proposed ZNF699–RNA-Pol-II interaction and ribosome-biogenesis hypothesis (polysome profiling, nascent-transcription assays) in patient/model hematopoietic cells; compare to *MYSM1*-deficient models.
5. **Build a natural-history registry:** aggregate reported and new patients to quantify penetrance, survival, genotype–phenotype correlations, and modifier candidates; formally establish prevalence and ontology mappings (Orphanet, ICD-11).
6. **Deploy the episignature clinically:** promote routine methylation-episignature testing to resolve *ZNF699* VUS and correct misdiagnoses, and to detect additional cases in existing methylation-array datasets.

Report compiled from 9 confirmed findings and 18 reviewed papers across 5 investigation iterations. Evidence types: predominantly human clinical (cohort + case reports), with in-silico structural modeling (AlphaFold2) and extrapolated in-vitro/model-organism mechanistic context from the broader KRAB-ZFP/KAP1 literature.
